

Let us Learn Scilab

Scilab is a FOSS programming environment that aids in numerical computations for engineering and scientific purposes. This tool is used for learning engineering sciences and automatic control engineering

ANAGHA P.

Scilab is a powerful computational workspace and programming language that comes with a rich set of algorithms and other features for various scientific applications. It is a free and open source software (FOSS) project licenced under CeCILL. It is distributed with the source code, and is also available for download in 32- and 64-bit executable versions for GNU/Linux, Windows and Mac OS.

Features

Scilab comes with a large number of functionalities that assist computations. Some of the features of this software are given below:

Mathematics and simulation. Scilab provides 1700 functions that help mathematical operations and data analyses in general science and engineering applications. In addition to the elementary mathematical functions, they can process linear algebra, sparse matrices, polynomials and rational functions. It can also perform various simulations (linear and non-linear differential equation solver).

Optimisation. Scilab provides algorithms to solve constrained and unconstrained continuous and discrete problems for large- and small-scale issues such as optimisations (linear, quadratic, non-linear and linear matrix inequality), semi-definite programming, genetic algorithms and simulated annealing.

Statistics. Data analysis and modelling tools are available for descriptive statistics, probability distributions and linear and non-linear modelling.

Control systems study. Control

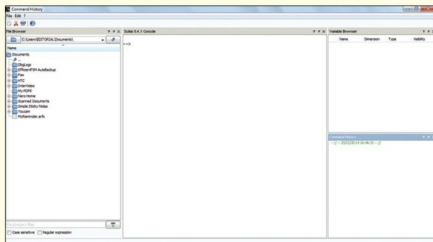


Fig. 1: Scilab main window

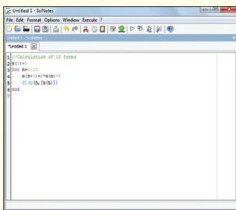
system design and analysis tools are transfer function, classic and robust control, stability analysis and state-space control system design.

Signal processing. It can perform signal generation, data windowing, power spectral density estimation, analogue filter design, digital FIR and IIR filter design and several signal transforms including Fastest Fourier Transform in the West (FFTW).

Customisation of graphics. Fig. 2: SciNotes

There are options provided for 2D and 3D visualisation. You can plot line charts, pie charts and histograms and animate them. These graphics can be saved and exported in PNG, PPM, EMF, EPS, FIG, PDF and SVG formats. The user can also add LaTeX or MathML annotations to it.

Xcos. It is a hybrid, dynamic system modeller and simulator in Scilab. It lets you design, save, load, compile and simulate models of mechanical systems, hydraulic circuits and control systems.



Application development. API enabling is provided to increase the native functionalities. Scilab has tools for creating and deploying user-defined modules. It also allows users to interface with external tools such as Matlab and Excel.

Installation

Source form of Scilab is available, which lets the users compile and produce their own binary format. Binary files for Linux, Windows and Mac OS